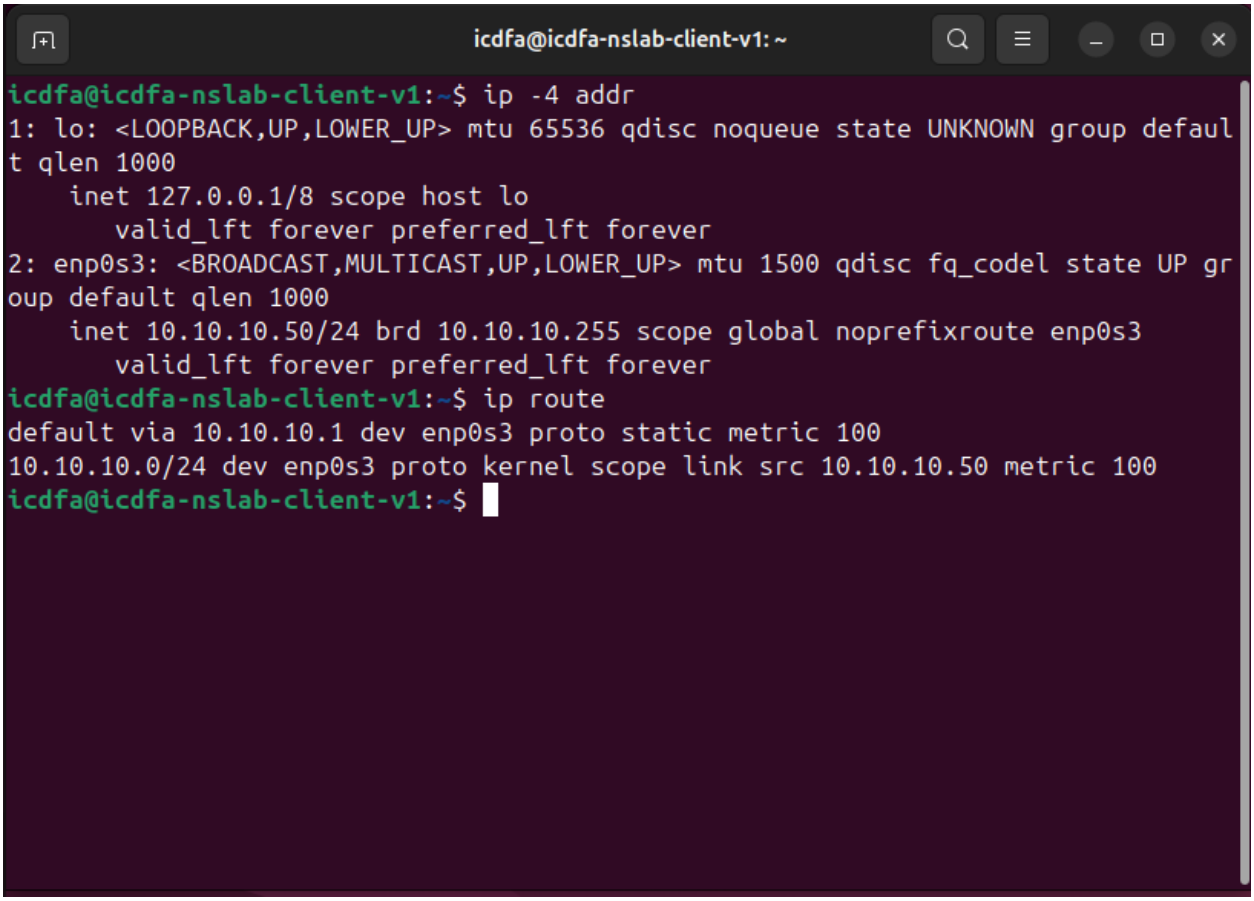
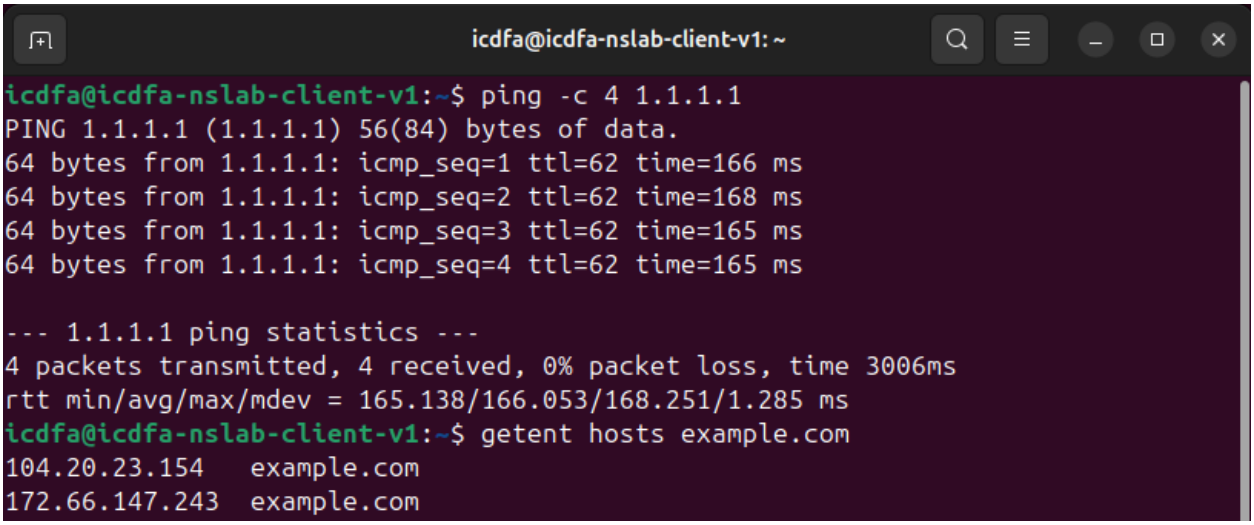
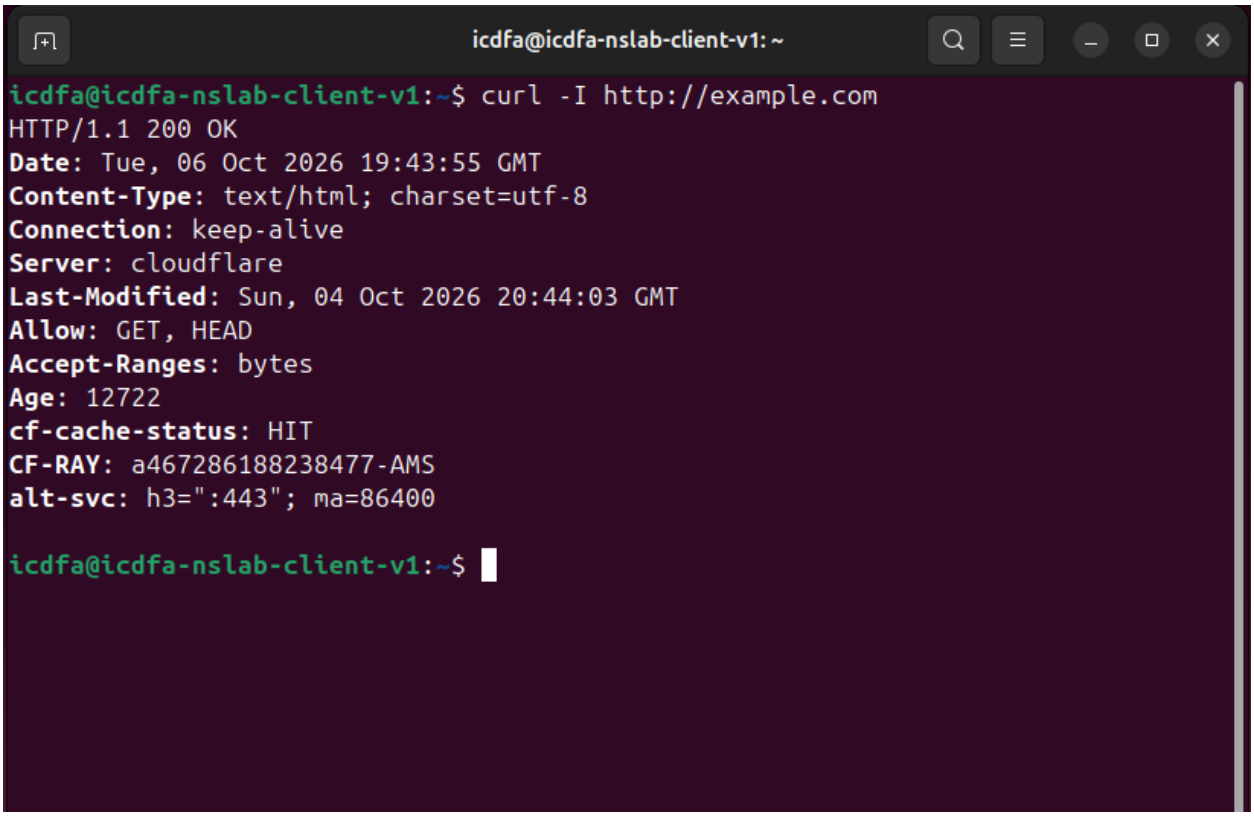
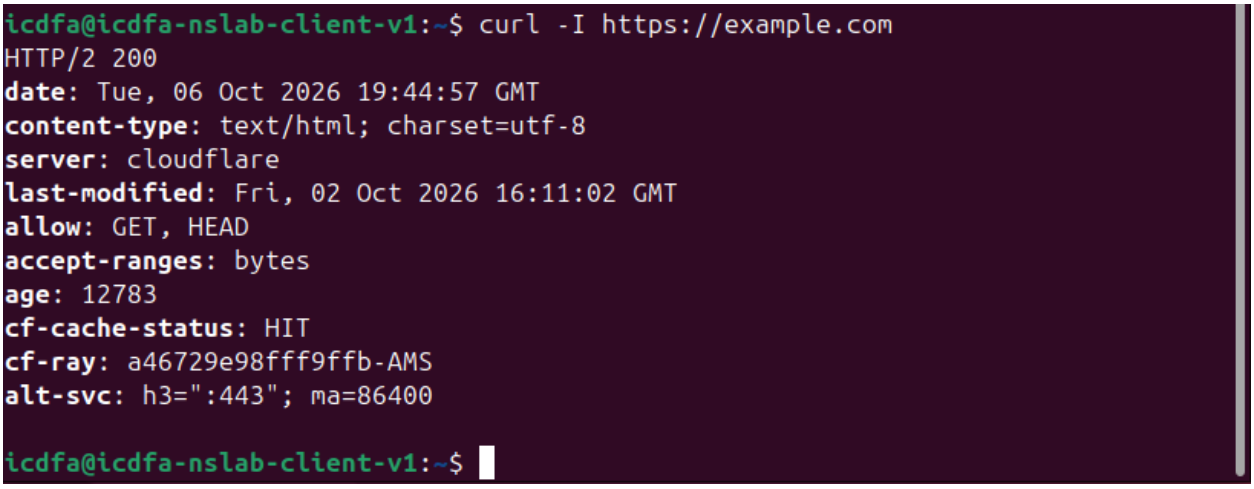
Full name: Adeyemo Adetoye
Registration number: C11/26/FCDF/17140
Linux distribution / version: Version 7.2.10r174163
VM hostname: Kali






icdfa@icdfa-nslab-client-v1: ~
icdfa@icdfa-nslab-client-v1:~$ curl -I http://example.com
HTTP/1.1 200 OK
Date: Tue, 06 Oct 2026 19:43:55 GMT
Content-Type: text/html; charset=utf-8
Connection: keep-alive
Server: cloudflare
Last-Modified: Sun, 04 Oct 2026 20:44:03 GMT
Allow: GET, HEAD
Accept-Ranges: bytes
Age: 12722
cf-cache-status: HIT
CF-RAY: a467286188238477-AMS
alt-svc: h3=":443"; ma=86400
icdfa@icdfa-nslab-client-v1:~$


icdfa@icdfa-nslab-client-v1:~$ curl -I https://example.com
HTTP/2 200
date: Tue, 06 Oct 2026 19:44:57 GMT
content-type: text/html; charset=utf-8
server: cloudflare
last-modified: Fri, 02 Oct 2026 16:11:02 GMT
allow: GET, HEAD
accept-ranges: bytes
age: 12783
cf-cache-status: HIT
cf-ray: a46729e98fff9ffb-AMS
alt-svc: h3=":443"; ma=86400
icdfa@icdfa-nslab-client-v1:~$

Icons		Version	Protocol	Source	Port	Destination	Port	Gateway	Descripti	Commands
	LAN	IPv4	ICMP	LAN network	*	1.1.1.1/32	*	*	LAB2 BLO ICMP TO 1.1.1.1	

```
icdfa@icdfa-nslab-client-v1: ~
icdfa@icdfa-nslab-client-v1:~$ ping -c 4 1.1.1.1
PING 1.1.1.1 (1.1.1.1) 56(84) bytes of data.

--- 1.1.1.1 ping statistics ---
4 packets transmitted, 0 received, 100% packet loss, time 3076ms

icdfa@icdfa-nslab-client-v1:~$
```

```
icdfa@icdfa-nslab-client-v1:~$ getent hosts example.com
2606:4700:10::6814:179a example.com
2606:4700:10::ac42:93f3 example.com
icdfa@icdfa-nslab-client-v1:~$
```

```
icdfa@icdfa-nslab-client-v1: ~
icdfa@icdfa-nslab-client-v1:~$ curl -I https://example.com
HTTP/2 200
date: Tue, 06 Oct 2026 20:25:11 GMT
content-type: text/html; charset=utf-8
server: cloudflare
last-modified: Fri, 02 Oct 2026 16:11:02 GMT
allow: GET, HEAD
accept-ranges: bytes
age: 796
cf-cache-status: HIT
cf-ray: a46764d41eaa971d-AMS
alt-svc: h3=":443"; ma=86400

icdfa@icdfa-nslab-client-v1:~$
```



Interface		Time	Protocol	Source	Destination	Action	Label
LAN	In	2026-10-06T22:08:30	TCP	10.10.10.50:…	91.189.91.5…	block	LAB2 BLOCK OUTBOUND HTTP
LAN	In	2026-10-06T22:08:22	TCP	10.10.10.50:…	91.189.91.5…	block	LAB2 BLOCK OUTBOUND HTTP
LAN	In	2026-10-06T22:08:18	TCP	10.10.10.50:…	91.189.91.5…	block	LAB2 BLOCK OUTBOUND HTTP
LAN	In	2026-10-06T22:08:16	TCP	10.10.10.50:…	91.189.91.5…	block	LAB2 BLOCK OUTBOUND HTTP

Interface		Time	Protocol	Source	Destination	Action	Label
LAN	In	2026-10-06T22:08:11	TCP	10.10.10.50:…	91.189.91.5…	block	LAB2 BLOCK OUTBOUND HTTP
LAN	In	2026-10-06T22:08:10	TCP	10.10.10.50:…	91.189.91.5…	block	LAB2 BLOCK OUTBOUND HTTP
LAN	In	2026-10-06T22:04:50	TCP	10.10.10.50:…	172.66.147.…	block	LAB2 BLOCK OUTBOUND HTTP
LAN	In	2026-10-06T22:04:49	TCP	10.10.10.50:	172.66.147	block	LAB2 BLOCK OUTBOUND HTTP

Interface		Time	Protocol	Source	Destination	Action	Label
LAN	In	2026-10-06T22:04:01	ICMP	10.10.10.50	1.1.1.1	block	LAB2 BLOCK ICMP TO 1.1.1.1
LAN	In	2026-10-06T22:04:00	ICMP	10.10.10.50	1.1.1.1	block	LAB2 BLOCK ICMP TO 1.1.1.1
LAN	In	2026-10-06T22:03:30	TCP	10.10.10.50:…	91.189.91.5…	block	LAB2 BLOCK OUTBOUND HTTP
LAN	In	2026-10-06T22:03:22	TCP	10.10.10.50:…	91.189.91.5…	block	LAB2 BLOCK OUTBOUND HTTP

Explain why the traffic matched the block rule instead of the default allow rule

The reason is the rule order:

1. LAB2 BLOCK OUTBOUND HTTP

2. LAB2 BLOCK ICMP TO 1.1.1.1

3. Default allow LAN to any

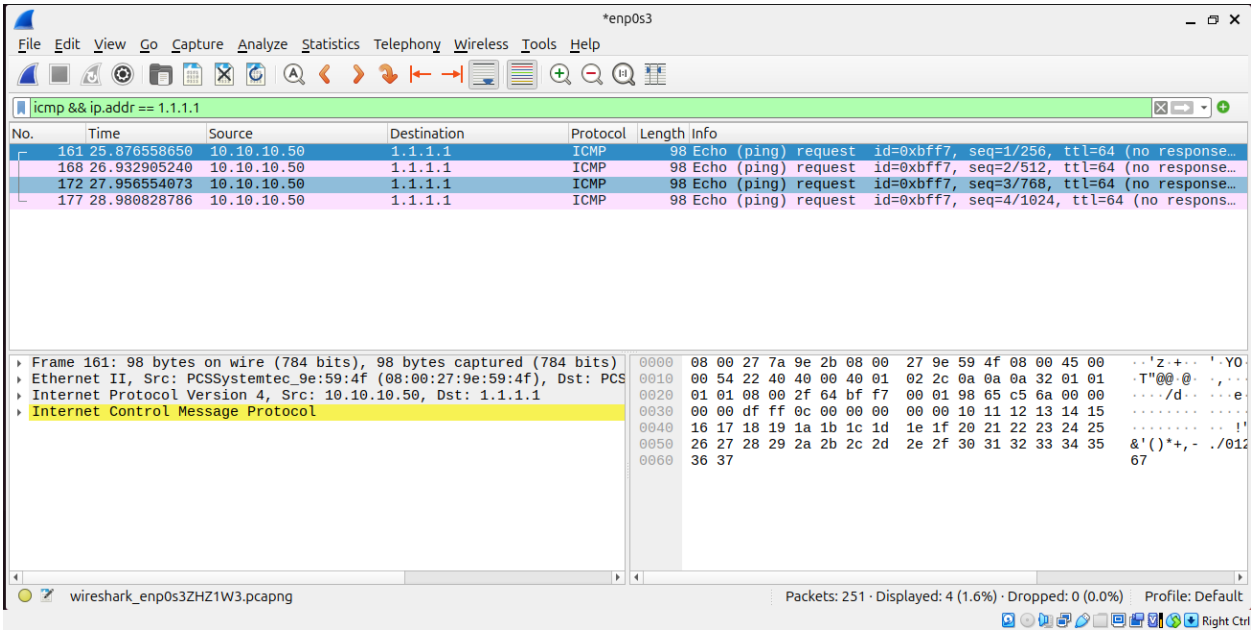
LAB2 BLOCK ICMP TO 1.1.1.1

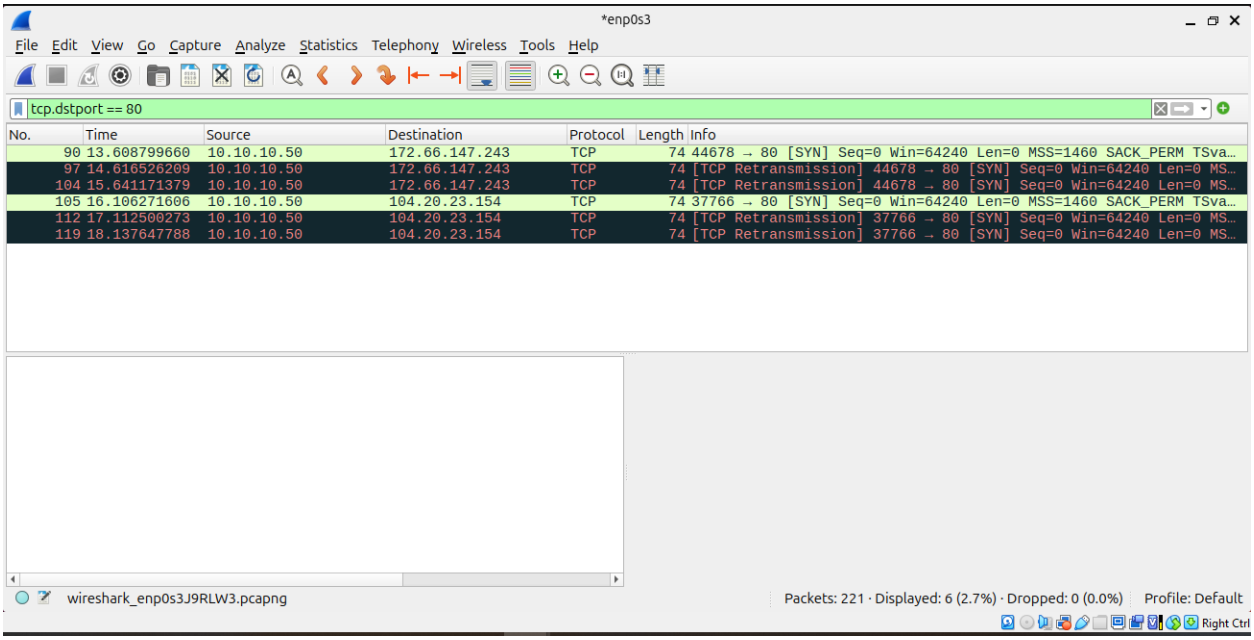
Therefore OPNsense blocks it before the packet can reach the broad default allow rule.

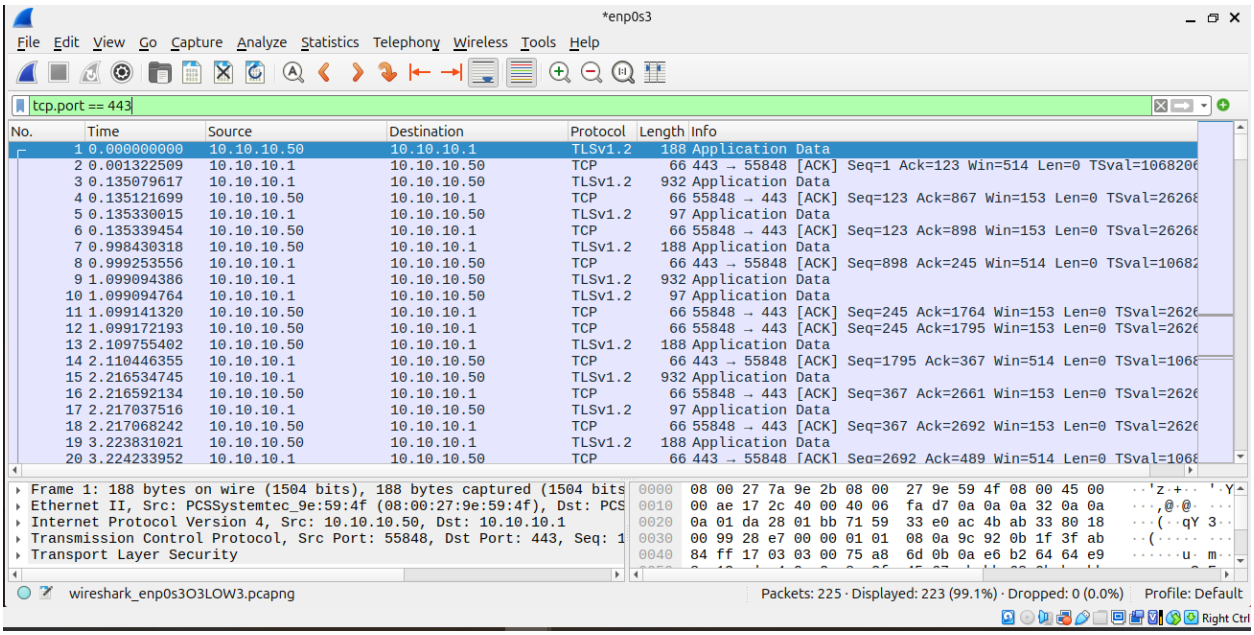
Likewise, an HTTP connection to TCP port 80 matches:

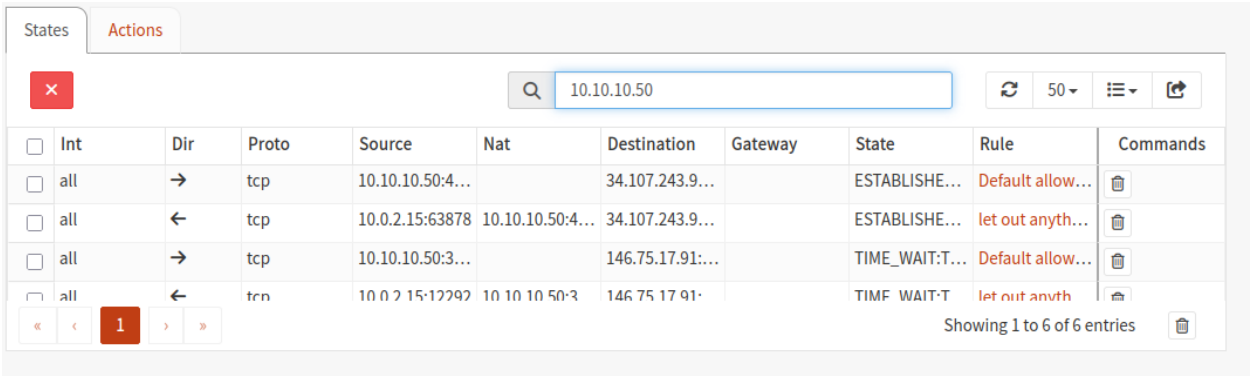
LAB2 BLOCK OUTBOUND HTTP

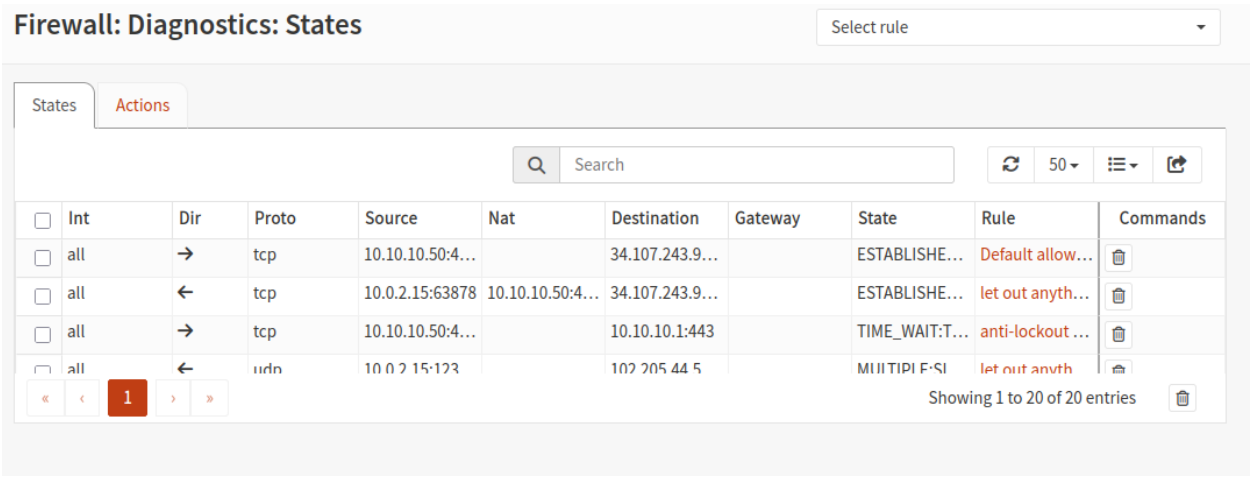
before reaching the default allow rule.



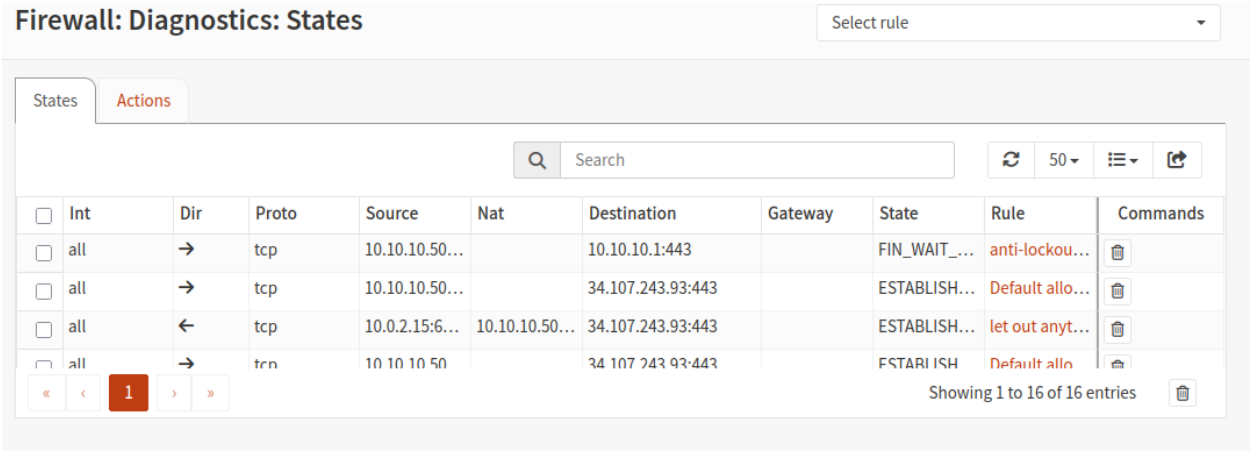

*enp0s3
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
tcp.dstport == 80
No. Time Source Destination Protocol Length Info
90 13.608799660 10.10.10.50 172.66.147.243 TCP 74 44678 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSva…
97 14.616526209 10.10.10.50 172.66.147.243 TCP 74 [TCP Retransmission] 44678 → 80 [SYN] Seq=0 Win=64240 Len=0 MS…
104 15.641171379 10.10.10.50 172.66.147.243 TCP 74 [TCP Retransmission] 44678 → 80 [SYN] Seq=0 Win=64240 Len=0 MS…
105 16.106271606 10.10.10.50 104.20.23.154 TCP 74 37766 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSva…
112 17.112500273 10.10.10.50 104.20.23.154 TCP 74 [TCP Retransmission] 37766 → 80 [SYN] Seq=0 Win=64240 Len=0 MS…
119 18.137647788 10.10.10.50 104.20.23.154 TCP 74 [TCP Retransmission] 37766 → 80 [SYN] Seq=0 Win=64240 Len=0 MS…
wireshark_enp0s3J9RLW3.pcapng
Packets: 221 · Displayed: 6 (2.7%) · Dropped: 0 (0.0%)
Profile: Default
Right Ctrl


*enp0s3
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
tcp.port == 443
No. Time Source Destination Protocol Length Info
1 0.000000000 10.10.10.50 10.10.10.1 TLSv1.2 188 Application Data
2 0.001322509 10.10.10.1 10.10.10.50 TCP 66 443 → 55848 [ACK] Seq=1 Ack=123 Win=514 Len=0 TSval=106820
3 0.135079617 10.10.10.1 10.10.10.50 TLSv1.2 932 Application Data
4 0.135121699 10.10.10.50 10.10.10.1 TCP 66 55848 → 443 [ACK] Seq=123 Ack=867 Win=153 Len=0 TSval=26268
5 0.135330015 10.10.10.1 10.10.10.50 TLSv1.2 97 Application Data
6 0.135339454 10.10.10.50 10.10.10.1 TCP 66 55848 → 443 [ACK] Seq=123 Ack=898 Win=153 Len=0 TSval=26268
7 0.998430318 10.10.10.50 10.10.10.1 TLSv1.2 188 Application Data
8 0.999253556 10.10.10.1 10.10.10.50 TCP 66 443 → 55848 [ACK] Seq=898 Ack=245 Win=514 Len=0 TSval=10682
9 1.099094386 10.10.10.1 10.10.10.50 TLSv1.2 932 Application Data
10 1.099094764 10.10.10.1 10.10.10.50 TLSv1.2 97 Application Data
11 1.099141320 10.10.10.50 10.10.10.1 TCP 66 55848 → 443 [ACK] Seq=245 Ack=1764 Win=153 Len=0 TSval=2626
12 1.099172193 10.10.10.50 10.10.10.1 TCP 66 55848 → 443 [ACK] Seq=245 Ack=1795 Win=153 Len=0 TSval=2626
13 2.109755402 10.10.10.50 10.10.10.1 TLSv1.2 188 Application Data
14 2.110446355 10.10.10.1 10.10.10.50 TCP 66 443 → 55848 [ACK] Seq=1795 Ack=367 Win=514 Len=0 TSval=1068
15 2.216534745 10.10.10.1 10.10.10.50 TLSv1.2 932 Application Data
16 2.216592134 10.10.10.50 10.10.10.1 TCP 66 55848 → 443 [ACK] Seq=367 Ack=2661 Win=153 Len=0 TSval=2626
17 2.217037516 10.10.10.1 10.10.10.50 TLSv1.2 97 Application Data
18 2.217068242 10.10.10.50 10.10.10.1 TCP 66 55848 → 443 [ACK] Seq=367 Ack=2692 Win=153 Len=0 TSval=2626
19 3.223831021 10.10.10.50 10.10.10.1 TLSv1.2 188 Application Data
20 3.224233952 10.10.10.1 10.10.10.50 TCP 66 443 → 55848 [ACK] Seq=2692 Ack=489 Win=514 Len=0 TSval=1068
Frame 1: 188 bytes on wire (1504 bits), 188 bytes captured (1504 bits
Ethernet II, Src: PCSSystemtec_9e:59:4f (08:00:27:9e:59:4f), Dst: PCS
Internet Protocol Version 4, Src: 10.10.10.50, Dst: 10.10.10.1
Transmission Control Protocol, Src Port: 55848, Dst Port: 443, Seq: 1
Transport Layer Security
0000 08 00 27 7a 9e 2b 08 00 27 9e 59 4f 08 00 45 00 ··'z·+·· '·YO··E·
0010 00 ae 17 2c 40 00 40 06 fa d7 0a 0a 0a 32 0a 0a ···,@·@· ·····2··
0020 0a 01 da 28 01 bb 71 59 33 e0 ac 4b ab 33 80 18 ···(··qY 3··K·3··
0030 00 99 28 e7 00 00 01 01 08 0a 9c 92 0b 1f 3f ab ··(····· ·····?·
0040 84 ff 17 03 03 00 75 a8 6d 0b 0a e6 b2 64 64 e9 ·····u· m····dd·
wireshark_enp0s3O3LOW3.pcapng
Packets: 225 · Displayed: 223 (99.1%) · Dropped: 0 (0.0%)
Profile: Default
Right Ctrl





LAN side of the state



Outbound NAT